

ALGEBRAIC STRUCTURES

Introduction

Let A be any set. A Mapping $f : A \times A \times \dots \times A \rightarrow A$ (or) $f : A^n \rightarrow A$ is called an n -ary operation and 'n' called the order of the operation.

Unary operation:

$$n=1, f : A \rightarrow A$$

Binary operation:

$$n=2, f : A \times A \rightarrow A$$

Algebraic system (or) algebra:

A set together with a number of operations (binary) on the set is called an **Algebraic system**

Example

- Semi groups
- Monoids
- Groups are algebraic systems with one binary operation.
- Rings, Integral domains ,Fields are algebraic systems with two binary operation.

Properties of Binary operations

- **Closure property:** A binary operation $*$: $G \times G \rightarrow G$ is said to be closed if for all $a, b \in G$ an element $a*b = x \in G$.
- **Associative:** $a*(b*c) = (a*b)*c$, for all $a, b, c \in G$
- **Existence of Identity:** There exists an element $e \in G$ such that $e*a = a*e = a$, $a \in G$ for all. The elements e called the identity element.
- **Existence of inverse:** For $a \in G$, there exists an element $b \in G$ such that $a*b = b*a = e$. The element b is called the inverse of 'a' and it is denoted by $b = a^{-1}$
- **Commutative:** For all $a, b \in G$, if $a*b = b*a$, then $*$ is commutative(abelian)
- **Distributive properties:**
 - $a*(b . c) = (a*b) . (a*c)$ (left distributive law)
 - $(b . c)*a = (b*a) . (c*a)$ (right distributive law)for all $a, b, c \in G$
- **Cancellation properties:**
 - $a*b = a*c \Rightarrow b = c$ [left cancellation Law]
 - $b*a = c*a \Rightarrow b = c$, for all $a, b, c \in G$ [right cancellation Law]

Semi groups:

- **Definition:**

A non-empty set S together with the binary operation $*$: $S \times S \rightarrow S$ is said to be a **Semi group** if $*$ satisfies the following conditions, namely the closure property and associative property. We denote the semi group by $(S, *)$.

- **Example.1**

Let $N = \{1, 2, 3, \dots\}$ be the set of natural numbers. Then $(N, +)$ and $(N, \times)$ are semi group under the binary operations addition and multiplication respectively.

- **Example.2**

Let $E = \{2, 4, 6, \dots\}$ be the set of all even numbers. Then $(E, +)$ and $(E, \times)$ are semi groups.

- Morphism of semi groups:**

Let $(S, *)$ and (T, Δ) be any two semi groups. A mapping $f : S \rightarrow T$ is said to be a **semi group homomorphism** if $f(a * b) = f(a) \Delta f(b)$, for all $a, b \in S$.

$$f : (S, *) \xrightarrow[\text{1-1}]{\text{homomorphism}} (T, \Delta) \rightarrow \text{monomorphism}$$

$$f : (S, *) \xrightarrow[\text{Onto}]{\text{homomorphism}} (T, \Delta) \rightarrow \text{epimorphism}$$

$$f : (S, *) \xrightarrow[\text{1-1 and onto}]{\text{homomorphism}} (T, \Delta) \rightarrow \text{isomorphism}$$

$$f : (S, *) \xrightarrow[\text{1-1}]{\text{isomorphism}} (T, \Delta) \rightarrow \text{endomorphism}$$

$$f : (S, *) \xrightarrow[\text{onto}]{\text{isomorphism}} (T, \Delta) \rightarrow \text{automorphism}$$

Example:

Let $(N, +)$ and $(Z_e, +_m)$ be any two semi groups.

Define a map $f : N \rightarrow Z_m$ by $f(a) = [a]_m$, for all $a \in N$.

Then $f(a + b) = [a + b]_m = [a] +_m [b] = f(a) +_m f(b)$

Therefore f is a semi group homomorphism.

Monoids:

- **Definition:**

A semi group $(M, *)$ with an identity element with respect to the operation $*$ is called a **Monoid (or)** a non empty set M together with the binary operation $*: M \times M \rightarrow M$ is said to be a **Monoid** if $*$ satisfies the closure property, associative property and identity property.

- **Example.1**

Let $N = \{1, 2, 3, \dots\}$ be the set of natural numbers. Then $(N, \times)$ is a monoid with the identity element **1**, but $(N, +)$ is not a monoid since the additive identity **0** is not a natural number.

- **Example.2**

Let $Z_+ = \{0, 1, 2, 3, \dots\}$ be the set of all non-negative integers. Then $(Z_+, +)$ and $(Z_+, \times)$ are semi groups as well as monoids.

- **Example.3**

Let S be a nonempty set and $P(S)$ be its power set. Then $(P(S), \cup)$ and $(P(S), \cap)$ are monoids with identities $\varnothing$ and S respectively.

- **Morphism of monoids:**

Let $(M, *, e_M)$ and (T, Δ, e_T) be any two monoids with the identity elements e_M and e_T respectively. A mapping

$f : M \rightarrow T$ is said to be a **monoid morphism** if for any two elements $a, b, \in M$, $f(a*b) = f(a) \Delta f(b)$ and $f(e_M) = e_T$

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$f : (M, *, e_M) \xrightarrow{1-1} (T, \Delta, e_T) \rightarrow \text{monomorphism}$

$f : (M, *, e_M) \xrightarrow{\text{onto}} (T, \Delta, e_T) \rightarrow \text{epimorphism}$

$f : (M, *, e_M) \xrightarrow{1-1 \text{ and onto}} (T, \Delta, e_T) \rightarrow \text{isomorphism}$

• **Theorem 1:**

For any commutative monoid $(M, *)$, the set of idempotent element of M forms a submonoid.

Proof:

Let $(M, *)$ be a commutative monoid.

Let $S = \{a \in M / a * a = a\}$, the set of idempotent elements of M .

Clearly $e \in S$, as $e * e = e$.

Let $a, b \in S$ with $a * a = a$ $b * b = b$.

Now $(a * b) * (a * b) = (a * b) * (b * a)$

$$\begin{aligned} &= a * (b * b) * a \\ &= a * (b) * a \\ &= a * (b * a) \\ &= a * (a * b) \\ &= (a * a) * b \\ &= a * b \end{aligned}$$

Hence $a * b \in S$

Therefore $(S, *)$ is a sub monoid of $(M, *)$

GROUPS

Definition (1):

A non-empty set G together with the binary operation $*$, $(G, *)$ is called a **group** if $*$ satisfies the following conditions

- **Associative:** For every $a, b, c \in G$, $a*(b*c)=(a*b)*c$.
- **Identity:** There exists an element $e \in G$ called the identity element such that $a*e=e*a=a$, for all $a \in G$.
- **Inverse:** There exists an element $a^{-1} \in G$ called the inverse of 'a' such that $a*a^{-1}=a^{-1}*a=e$, for each $a \in G$

Definition (2):

A group $(G, *)$ is said to be abelian group if $a*b=b*a$, for all $a, b \in G$. otherwise it is non-abelian group.

Definition (3):

The order of a group $(G, *)$, denote by $O(G)$ or $|G|$, is the number of elements of G , is finite.

Example (1):

The set of all integers $\mathbb{Z}$ with the operation of ordinary addition is an infinite abelian group.

Example (2):

The set of all integers $\mathbb{Z}$ is not a group under multiplication. $(\mathbb{Z}, \cdot)$ is not a group. Because there is no multiplicative inverse in $\mathbb{Z}$. But $(\mathbb{Z}, \cdot)$ is a monoid and hence a semi group.

Example (3):

The set of all non-zero real numbers $\mathbb{R}$ form an infinite abelian group under the binary operation $*$ defined by $a*b=ab/2$, for all $a, b \in \mathbb{R}$

Example (4):

The set of all non-zero real numbers $\mathbb{R}$, under the operation of ordinary multiplication is a group. An inverse of $a \neq 0$ is $1/a$ i.e. $a^{-1} = 1/a$

Example (5):

Prove that the set $A=\{1, \omega, \omega^2\}$ is an abelian group of order 3 under multiplication. Where $1, \omega, \omega^2$ are cube roots of unity and $\omega^3=1$.

Properties of groups

Properties of groups

Property 1:

The identity of a group is unique (or) If $(G, *)$ is a group and e is an identity of G , then no other element of G is an identity of G .

Proof:

Suppose that e_1 and e_2 are two identities of the group $(G, *)$.

Now e_1 is the identity, then $e_1 * e_2 = e_2 * e_1 = e_2 \rightarrow (1)$

Again e_2 is the identity, then $e_2 * e_1 = e_1 * e_2 = e_1 \rightarrow (2)$

From (1) and (2) we see that $e_1 = e_2$

The identity is unique

Property 2:

In a group, the left and right cancellation laws are true. That $a*b=a*c \Rightarrow b=c$ and $b*a=c*a \Rightarrow b=c$.

Proof:

Let $G=(G, *)$ be a group For $a \in G$, there exists an $a^{-1} \in G$ such that $a*a^{-1}=a^{-1}*a=e$.

(i)Left cancellation law:

Let $a*b=a*c$.

$$\begin{aligned} a^{-1}*(a*b) &= a^{-1}*(a*c) \Rightarrow (a^{-1}*a)*b = (a^{-1}*a)*c \\ &\Rightarrow e*b = e*c \Rightarrow b=c. \end{aligned}$$

(ii)Right cancellation law :

$$\begin{aligned} \Rightarrow \text{Let } b*a &= c*a \Rightarrow (b*a)*a^{-1} = (c*a)*a^{-1} \\ &\Rightarrow b*(a*a^{-1}) = c*(a*a^{-1}) \\ &\Rightarrow b*e = c*e \Rightarrow b=c. \end{aligned}$$

Property 3:

The inverse of any element of group is unique.

Proof:

Let G be a group, $a \in G$ and e be the identity element of G .

Suppose that b and c be two inverses of an element $a \in G$.

$$b * a = e = a * b \quad \text{and} \quad c * a = e = a * c.$$

$$\text{Now } (b * a) * c = c \Rightarrow b * (a * c) = c$$

$$\Rightarrow b * e = c \Rightarrow b = c$$

There is one and only one inverse for each element in a group.

Property 4:

A group can not have any element which is idempotent except the identity element

Proof:

Let G be a group. Let us assume that $a \in G$ is idempotent. Then

$$a * a = a.$$

Now
$$e = a^{-1} * a = a^{-1} * (a * a) = (a^{-1} * a) * a$$

$$= e * a = a$$

$$\Rightarrow e = a$$

Property 5: If a is an element of group G , then $(a^{-1})^{-1} = a$ (or) If the inverse of a is a^{-1} , then the inverse of a^{-1} is a .

Proof:

If e is the identity of the group G , then

$a^{-1} * a = e = a * a^{-1}$, for every $a \in G$.

$$(a^{-1})^{-1} * (a^{-1} * a) = (a^{-1})^{-1} * e = (a^{-1})^{-1}$$

$$\Rightarrow (a^{-1})^{-1} * (a^{-1} * a) = (a^{-1})^{-1}$$

$$\Rightarrow ((a^{-1})^{-1} * a^{-1}) * a = (a^{-1})^{-1}$$

$$\Rightarrow e * a = (a^{-1})^{-1}$$

$$\Rightarrow a = (a^{-1})^{-1}$$

Property 6:

If a has inverse b and b has inverse c , then $a=c$.

Proof:

$$\text{Given } a*b = e = b*a \quad \rightarrow (1)$$

$$\text{and } b*c = e = c*b \quad \rightarrow (2)$$

$$a = a*e = a*(b*c)$$

$$= e*c$$

$$= c$$

$$a = c$$

Property 7:

The inverse of the product of two element of a group is the product of their inverse order. That is $(a * b)^{-1} = b^{-1} * a^{-1}$.

Proof:

Let $a, b \in G$ and a^{-1}, b^{-1} be their inverse respectively .

$$a * a^{-1} = e = a^{-1} * a \text{ and } b * b^{-1} = e = b^{-1} * b$$

$$\begin{aligned} \text{Now } (a * b) * (b^{-1} * a^{-1}) &= a * [b * (b^{-1} * a^{-1})] \\ &= a * [(b * b^{-1}) * a^{-1}] \\ &= a * [e * a^{-1}] = a * a^{-1} \\ &= e \end{aligned}$$

Similarly we can prove $(b^{-1} * a^{-1}) * (a * b) = e$.

$$(a * b) * (b^{-1} * a^{-1}) = e = (b^{-1} * a^{-1}) * (a * b).$$

We see that $(b^{-1} * a^{-1})$ is the inverse of $(a * b)$

That is $(a * b)^{-1}$ is the inverse of $(a * b)$

$$\text{That is } (a * b)^{-1} = b^{-1} * a^{-1}$$

Property 8:

There existence of a unique inverse of every element of G guarantees the unique solvability of any equation of the type $a*x = b$ has a unique solution $x = a^{-1}*b$. in addition, the equation $x*a=b$ has a unique solution $x = b* a^{-1}$.

Proof:

Let G be a group. The for any $a, b \in G$,
we have $a* a^{-1} = e = a^{-1}*a$ and $b*b^{-1} = e = b^{-1}*b$.

$$\begin{aligned}\text{Now} \quad a * x &= b = e * b \\ &= (a*a^{-1})*b \\ &= a * (a^{-1}*b) \\ \Rightarrow x &= a^{-1}*b\end{aligned}$$

To prove this solution is unique, let us suppose x_1 and x_2 are two solutions of $a* x = b$.
Then

$$\begin{aligned}a * x_1 &= b \text{ and } a*x_2=b \\ a* x_1 &= a* x_2 \\ \Rightarrow x_1 &= x_2\end{aligned}$$

$x = a^{-1}* b$ is the unique solution for $a* x = b$. similarly we can prove the second part.

Property 9:

A group G is abelian iff $(a * b)^2 = a^2 * b^2$

Proof:

Let us assume that G is abelian. Hence, for $a, b \in G$.

We have $a * b = b * a$

Now

$$\begin{aligned} a^2 * b^2 &= (a * a) * (b * b) \\ &= a * [a * (b * b)] \\ &= a * [(a * b) * b] \\ &= a * [(b * a) * b] \\ &= (a * b) * (a * b) \\ &= (a * b)^2 \end{aligned}$$

$$(a * b)^2 = a^2 * b^2$$

Conversely, assume that $(a * b)^2 = a^2 * b^2$

$$\Rightarrow (a * b) * (a * b) = (a * a) * (b * b)$$

$$\Rightarrow a * [b * (a * b)] = a * [a * (b * b)]$$

$$\Rightarrow b * (a * b) = a * (b * b)$$

$$\Rightarrow (b * a) * b = (a * b) * b$$

$$\Rightarrow b * a = a * b$$

$\Rightarrow G$ is abelian

Property 10:

For any group G , if $a^2 = e$ with $a \neq e$, then G is abelian

(Or)

If every element of group G is its own inverse, then G is abelian. Is the converse also true.

Proof:

Let G be a group and $a, b \in G$, then $a * b \in G$.

Given $a = a^{-1}$ and $b = b^{-1}$

$$(a * b) = (a * b)^{-1}$$

$$= b^{-1} * a^{-1}$$

$$= b * a$$

$$\Rightarrow G \text{ is abelian}$$

Modular Arithmetic

Modular Arithmetic:

- The addition modulo n is defined by $a +_n b = r$, where a, b are integers, n is a +ve integer and r is the least +ve remainder when $(a+b)$ is divided by n .
 $a +_n b$ = the least +ve remainder when $(a+b)$ is divided by n .

Eg: $8+_5 3 = 11 = 2(5)+1 = 1$ (Addition mod 5)

$-23+_7 3 = -20 = -3(7) + 1 = 1$ (Addition mod 7)

The multiplication modulo n is defined by $a \cdot b$ = the least +ve remainder when $(a \cdot b)$ is divided by n .

$3 \cdot 5 = 15 = 3(4) + 3 = 3$

$-5 \cdot 6 = -30 = -8(4) + 2 = 2$

Properties of Modular Arithmetic on Z_n

Addition modulo n is always commutative and associative. 0 is the identity for $+_n$ and every element of Z_n has an additive inverse .

Multiplication modulo n is always commutative and associative and 1 is identity for $\cdot_n$. multiplication modulo n is distributive over addition modulo n . also every element of Z_n has the multiplicative inverse.

Example 1:

Prove that $(\mathbb{Z}_3, +_3)$ is an abelian group.

Let $\mathbb{Z}_3 = \{0,1,2\}$, Let us defined the addition table as

$+_3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

For $x, y \in \mathbb{Z}_3$, define,

$x +_3 y =$ the least +ve reminder , when $(x+y)$ divided by 3,

since all the entries in the addition table are the element of $\mathbb{Z}_3$, it is closed under $+_3$. Associative, commutative, with identity element 0 and every element of $\mathbb{Z}_n$ has an additive inverse , we see that $(\mathbb{Z}_3, +_3)$ is an additive abelian group.

Example 2:

Prove that $(Z_5, \times_5)$ is an abelian group.

Solution:

Let $Z_5 = \{1,2,3,4\}$. Let us define the multiplication mod 5 table as

$\times_5$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Since all the entries in the multiplication table are the element of Z_5 , It is closed under $\times_5$. Here 1 is the identity. Also from fact that multiplication mod n is always associative, commutative and every element of Z_n has a multiplicative inverse, we see that $(Z_5, \times_5)$ is an abelian group.

Sub Groups

Definition:

A non-empty subset H of a group G is said to be a subgroup of G if H itself is a group under the same operation defined on G and with the same identity element.

Let $(G, *)$ be a group and $H \subseteq G$ ($H \neq \varnothing$) is said to be subgroup $(H, *)$ of $(G, *)$ if

1. $e \in H$, where e is the identity of G
2. For any $a \in H$, $a^{-1} \in H$
3. For $a, b \in H$, $a * b \in H$.

Example :

1. The set of all integers is a subgroup of the set of all real numbers under addition. That is $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{R}, +)$
2. The set of even integer $2\mathbb{Z} = \{2k/k \in \mathbb{Z}\}$ is a subgroup of $(\mathbb{Z}, +)$
3. The set of non-negative integers is not a subgroup of $(\mathbb{Z}, +)$, because, except 0, no element have the additive inverse.

Theorem 1

The necessary and sufficient condition that a non-empty subset H of a group G be a subgroup is $a \in H, b \in H \Rightarrow a * b^{-1} \in H$

Proof

Necessary Conditions:

Assume that H is a subgroup of G . since H itself is a group ; we have $a, b \in H$.
Also $b \in H \Rightarrow b^{-1} \in H$.
 $a, b \in H, b^{-1} \in H \Rightarrow a * b^{-1} \in H$

Sufficient Conditions : Let $a * b^{-1} \in H \rightarrow (1)$ for all $a, b \in H$ and $H \subseteq G$. we have to prove that H is a subgroup of G .

For,

Identity : Let $a \in H, a \in H \Rightarrow a * a^{-1} \in H$
 $\Rightarrow e \in H$

Hence the identity e is the element of H

Existence of inverse :

Let $e \in H, a \in H \Rightarrow e * a^{-1} \in H \Rightarrow a^{-1} \in H$
Every element of H has an inverse which is in H .

•Closure :

Let $b \in H \Rightarrow b^{-1} \in H$

For $a, b \in H \Rightarrow a, b^{-1} \in H \Rightarrow a * (b^{-1})^{-1} \in H$

$\Rightarrow a * b \in H$

H is closed under the composition.

Associative :

Since $H \subseteq G$, the elements of H are also the elements of G. since the composition in G is associative, it must also be associative in H.

H itself is a group for the composition in G.

H is a subgroup of G.

Theorem 2:

The intersection of two subgroups of a group is also a subgroup of the group (OR) If H_1, H_2 are two subgroups of a group G , then $H_1 \cap H_2$ is also a subgroup of G .

Proof:

Let H_1 and H_2 be any two subgroups of G then $H_1 \cap H_2 \neq \emptyset$, because at least the identity element is common to both H_1 and H_2 .

To prove $H_1 \cap H_2$ is a subgroup, it is enough if we show that, for $a, b \in H_1 \cap H_2 \Rightarrow a * b^{-1} \in H_1 \cap H_2 \Rightarrow a * b^{-1} \in H_1 \cap H_2$.

For let $a, b \in H_1 \cap H_2 \Rightarrow a, b \in H_1$ and $a, b \in H_2$

$\Rightarrow a * b^{-1} \in H_1$ and $a * b^{-1} \in H_2$

$\Rightarrow a * b^{-1} \in H_1 \cap H_2$

$a, b \in H_1 \cap H_2 \Rightarrow a * b^{-1} \in H_1 \cap H_2$

$\Rightarrow H_1 \cap H_2$ is a subgroup of G

Theorem 3:

The union of two subgroups of G need not be a subgroup of G .

Proof:

To prove this consider the following example. Let G be a group of integers under addition

$(\mathbb{Z}, +)$ is a group

Now define $H_1 = \{x \mid x = 2n, n \in \mathbb{Z}\} = \{0, \pm 2, \pm 4, \pm 6, \dots\}$

$H_2 = \{x \mid x = 3n, n \in \mathbb{Z}\} = \{0, \pm 3, \pm 6, \pm 9, \pm 12, \dots\}$

We see that H_1 and H_2 are subgroups of G .

Define $H_1 \cup H_2 = \{x \mid x \in H_1 \text{ or } x \in H_2\}$
 $= \{0, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 9, \dots\}$

For $2, 9 \in H_1 \cup H_2 \Rightarrow 2 + 9 = 11 \notin H_1 \cup H_2$

$H_1 \cup H_2$ is not closed under addition

$H_1 \cup H_2$ is not a group

$H_1 \cup H_2$ is not a subgroup of G

Cyclic Groups

Definition:

A group $(G, *)$ is said to be cyclic if there exists $a \in G$ such that any $x \in G$ can be written as either $x = a^n$ or $x = na$, where n is some integer. Here the element 'a' is called the 'generator' of the cyclic group G . That is the cyclic group generated by 'a' and we denote it by $G = \langle a \rangle$

Theorem 1:

Every cyclic group is abelian.

Proof:

Let $(G, *)$ be a cyclic group generated by an element $a \in G$. i.e $G = \langle a \rangle$. Then for any two elements $x, y \in G$, we have $x = a^n$, $y = a^m$, where m, n are integers.

$$\begin{aligned} x * y &= a^n * a^m = a^{n+m} = a^{m+n} \\ &= a^m * a^n \\ &= y * x \end{aligned}$$

$(G, *)$ is abelian.

Theorem 2:

If 'a' is a generator of a cyclic group G, then a^{-1} is also a generator of G (or). If a is an element of group G, then $(a) = (a^{-1})$.

Proof

Let $G = (a)$ be a cyclic group generated by 'a'. Then $a^r \in G$, where r is some integer we can write $a^r = (a^{-1})^{-r}$, since $-r$ is also some integer

Each element of G is generated by a^{-1}

a^{-1} is also a generator of G.

$(G) = (a) = (a^{-1})$.

Theorem 3:

Prove that the group $G = \{1, -1, i, -i\}$ is cyclic and find its generators.

Solution

We can write $1 = (i)^4$, $-1 = (i)^2$, $i = (i)^1$, $-i = (i)^3$

i.e, All the elements of G can be expressed as integral powers of the element i.

G is a cyclic group generated by i.

Since 'i' is the generator of G, $(i)^{-1}$ is also a generator of G. Hence G is a cyclic group and its generators are I and $(i)^{-1} = 1/i = -i$

Theorem 4:

Every subgroup of a cyclic group is cyclic

Theorem 5:

Let $(G, *)$ be a finite group generated by an element $a \in G$. If G is of order n , that is $O(G) = n$, then $a^n = e$ so that $G = \{a, a^2, \dots, a^n = e\}$. Further, n is the least positive integer for which $a^n = e$.

Theorem 6:

The direct product of two or more groups is again a group.

Morphism of Group

Definition:

Let $(G, *)$ and (H, Δ) be any two groups. A mapping $f: G \rightarrow H$ is said to be a **homomorphism** if $f(a * b) = f(a) \Delta f(b)$, for any $a, b \in G$.

Theorem 1:

If f is a homomorphism of a group G into a group G' , then

(i) Group homomorphism preserves identities

i.e., $f(e) = e'$, where e is the identity of G and e' is the identity of G'

(ii) (Group homomorphism preserves inverses

i.e., $f(a^{-1}) = [f(a)]^{-1}$, for all $a \in G$.

Proof:

Let $a \in G$ and f is a homomorphism from G into G' , then $f(a) \in G'$

$$\begin{aligned} f(a) * e' &= f(a) = f(a * e) \\ &= f(a) * f(e) \\ \Rightarrow e' &= f(e) \end{aligned}$$

Let $a \in G$, then $a^{-1} \in G$ and $a * a^{-1} = e = a^{-1} * a$

$$\begin{aligned} e' &= f(e) = f(a * a^{-1}) \\ &= f(a) * f(a^{-1}) \end{aligned}$$

$$\Rightarrow f(a) * f(a^{-1}) = e'$$

We see that $f(a^{-1})$ is the inverse of $f(a)$ in G'

$$\text{i.e., } [f(a)]^{-1} = f(a^{-1})$$

Theorem 2:

Let f be a homomorphism from G to G' . Let $f(G)$ be the homomorphic image of G into G' is a subgroup of G' .

Proof:

Let $f(G) = \{f(x) \mid x \in G\}$

Clearly $f(G)$ is a non-empty subset of G' .

Now for any $a', b' \in f(G)$, we have $f(a) = a', f(b) = b'$, for $a, b \in G$.

$$\begin{aligned} a' * (b')^{-1} &= f(a) * f(b)^{-1} \\ &= f(a) * f(b^{-1}) \\ &= f(a * b^{-1}) \end{aligned}$$

But $a * b^{-1} \in G \Rightarrow f(a * b^{-1}) \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

i.e., For $a', b' \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

$\Rightarrow f(G)$ is a subgroup of G' .

Theorem 3:

Let $f: G \rightarrow G'$ be a group homomorphism and H is a subgroup of G' , then $f^{-1}(H)$ is a subgroup of G .

Proof:

Let $f^{-1}(H) = \{ x = f^{-1}(y) \in G \mid f(x) = y \in H \}$

Since H is a subgroup of G' , the set $f^{-1}(H)$ will be a non-empty subset of G .

Now consider $x_1 = f^{-1}(y_1)$, $x_2 = f^{-1}(y_2) \in f^{-1}(H)$ for any $y_1, y_2 \in H$

With $f(x_1) = y_1$ and $f(x_2) = y_2$.

$$\begin{aligned} \text{Let } x_1, x_2 \in f^{-1}(H) &\Rightarrow f(x_1), f(x_2) \in H \\ &\Rightarrow f(x_1) * [f(x_2)]^{-1} \in H \\ &\Rightarrow f(x_1) * f(x_2^{-1}) \in H \\ &\Rightarrow f(x_1 * x_2^{-1}) \in H \\ &\Rightarrow x_1 * x_2^{-1} \in f^{-1}(H) \\ x_1, x_2 \in f^{-1}(H) &\Rightarrow x_1 * x_2^{-1} \in f^{-1}(H) \\ f^{-1}(H) &\text{ is a subgroup of } G. \end{aligned}$$

Kernel of a homomorphism:

Definition:

Let $f: G \rightarrow G'$ be a group homomorphism

The set of elements of G which are mapped into e' , the identity of G' is called the kernel of f and is denoted by $\text{Ker}(f)$

i.e., $\text{Ker}(f) = \{ x \in G / f(x) = e', \text{the identity of } G' \}$

Theorem:

Proof:

$\text{Ker}(f) = \{ x \in G / f(x) = e', \text{the identity of } G' \}$

Since $f(e) = e'$ is true always, at least $e \in \text{Ker}(f)$

i.e., $\text{Ker}(f)$ is a non-empty subset of G .

Let $a, b \in \text{Ker}(f)$, with $f(a) = e'$ and $f(b) = e'$

$$\begin{aligned} f(a * b^{-1}) &= f(a) * f(b^{-1}) \\ &= f(a) * (f(b))^{-1} \\ &= e' * e' \\ &= e' \end{aligned}$$

$$\Rightarrow a * b^{-1} \in \text{Ker}(f)$$

i.e., $a, b \in \text{Ker}(f) \Rightarrow a * b^{-1} \in \text{Ker}(f)$

$\text{Ker}(f)$ is a subgroup of G .

Definition:

A mapping f from a Group G to a group G' is said to be an **Isomorphism** if

- f is a homomorphism i.e. $f(a \cdot b) = f(a) \cdot f(b)$, for any $a, b \in G$
- f is one-one. (injective) i.e. distinct elements of G have different f -image in G' .
- f is onto (surjective) i.e. every element of G' should be the f -image of some elements in G .

Cosets

Definition:

- For any $a \in G$, the set $a * H$ defined by $a * H = \{a * h / h \in H\}$ is called the **Left coset** of H in G determined by the element $a \in G$.
- For any $a \in G$, the set $H * a$ defined by $H * a = \{h * a / h \in H\}$ is called the **Right coset** of H in G determined by the element $a \in G$.

The element $a \in G$ is called the representative element of the left coset $a * H$ and right coset $H * a$

Note

The right or left coset of H in G is not empty.

Since $e \in H$, $e * H = H = H * e$, H itself is right as well as left coset .

$H * a$, $a * H$ are also subsets of G

If G is abelian, then $a * H = H * a$

In all the subsequent discussion in this chapter, aH means $a * H$ and Ha means $H * a$

Example.1

Let $G = \{ 1, a, a^2, a^3 \}$ be a group and $H = \{1, a^2\}$ is a subgroup of G under multiplication. Find all the cosets of H .

Solution:

Let us find the right cosets of H in G .

$$H1 = \{ 1, a^2 \} = H$$

$$Ha = \{a, a^2\},$$

$$Ha^2 = \{a^2, a^4\} = \{a^2, 1\} = H$$

$$Ha^3 = \{a^3, a^5\} = \{a^3, a\} = Ha$$

$$H.1 = H = Ha^2 = \{1, a^2\} \text{ and } Ha = Ha^3 = \{a, a^3\}$$

are two distinct right cosets of H in G . Similarly we can find the left cosets of H in G .

Example.2

Find the right cosets of $\{[0],[2]\}$ in the group $(Z_4, +_4)$.

Solution

Let $Z_4 = \{[0],[1],[2],[3]\}$ be a group and $H = \{[0],[2]\}$ be a subgroup of Z_4 under $+_4$ (addition mode 4).

The left cosets of H are

$$[0] + H = \{[0],[2]\} = H ;$$

$$[1] + H = \{[1],[3]\} ;$$

$$[2] + H = \{[2],[4]\} = \{[2],[0]\} = \{[0],[2]\} = H$$

$$[3] + H = \{[3],[5]\} = \{[3],[1]\} = \{[1],[3]\} = [1] + H.$$

$$[0] + H = [2] + H = H \text{ and } [1] + H = [3] + H \text{ are the two distinct left cosets of } H \text{ in } Z_4.$$

Note:

The union of all distinct left cosets of H in G is equal to G .

If H any subgroup of G and $a \in H$, then $Ha = H = aH$.

Theorem1:

If $a \in Hb$ then $Ha = Hb$ and if $a \in Hb$, then $aH = bH$.

Theorem2:

Any two right cosets of H in G are either disjoint or identical.

Definition:

A subgroup $(H, *)$ of group $(G, *)$ is said to be a **normal subgroup** of G , if for every $x \in G$ and for every $h \in H$, $xhx^{-1} \in H$ $xHx^{-1} = H$.

Theorem1:

A subgroup H of a group G is normal iff $xHx^{-1} = H$

Proof:

Let $xHx^{-1} = H \Rightarrow xHx^{-1} \subseteq H$, for all $x \in G$
 $\Rightarrow H$ is a normal subgroup of G

Conversely,

assume that H is a normal subgroup of G , then $xHx^{-1} \subseteq H$, for all $x \in G \rightarrow (1)$

Now $x \in G \Rightarrow x^{-1} \in G$, then

$x^{-1}H(x^{-1})^{-1} \subseteq H$, for all $x \in G$

$\Rightarrow x^{-1}Hx \subseteq H \Rightarrow x(x^{-1}Hx) x^{-1} \subseteq xHx^{-1}$

$\Rightarrow x^{-1}xHx x^{-1} \subseteq xHx^{-1} \Rightarrow e H e \subseteq xHx^{-1}$

$\Rightarrow H \subseteq xHx^{-1}$, for all $x \in G \rightarrow (2)$

From (1) and (2)

$xHx^{-1} = H$, for all $x \in G$.

Definition:

Let $(H, *)$ be a subgroup of the group $(G, *)$. We define the congruency relation on G called a left cosets relation with respect to the subgroup H or a left cosets relation mod H denoted by $\sim$ such that for $a, b \in G$, **a is congruent to b mod H** written $a \sim b \pmod{H}$ iff $b^{-1} * a \in H$.

Theorem1:

The left cosets relation mod H on $(G, *)$ defined by, for $a, b \in G$, $a \sim b \pmod{H}$ iff $b^{-1} * a \in H$ is an equivalence relation.

Theorem2:

Let $(H, *)$ be a subgroup of $(G, *)$. The set of left cosets of H in G forms a partition of G .

Theorem3:

If $(H, *)$ is a subgroup of a group $(G, *)$ and right cosets of H in G , then there exists a one-to-one correspondence between the elements of H and Ha (or)

$$O(H) = O(Ha).$$

Note:

There is a one-to-one correspondence between H and aH .

$$O(H) = O(aH)$$

Lagrange's theorem

Theorem 1:

The order of each subgroup of a finite group is a divisor of the order of the group.

Proof:

Let $(G, *)$ be finite group and $O(G) = n$. Let $(H, *)$ be a subgroup of $(G, *)$ and $O(H) = m$

Suppose that $h_1, h_2, h_3, \dots, h_m$ are the m members of H . For $a \in G$, the right coset Ha of H in G is defined by

$$Ha = \{h_1 * a, h_2 * a, h_3 * a, \dots, h_m * a\}$$

Since there should be a one-to-one correspondence between H and Ha , the members of Ha are distinct.

Hence each right coset of H in G has m distinct members.

We know that any right cosets of H in G are either disjoint or identical. Since G is a finite group, the number of distinct right cosets of H in G will be finite. The union of these k distinct right cosets of H in G is equal to G .

Hence, if $Ha_1, Ha_2, Ha_3, \dots, Ha_k$ are the k distinct right cosets of H in G , then

$$G = Ha_1 \cup Ha_2 \cup Ha_3 \cup \dots \cup Ha_k$$

$$O(G) = O(Ha_1) + O(Ha_2) + O(Ha_3) + \dots + O(Ha_k)$$

$$n = m + m + m + \dots + m$$

(k times)

$$n = km \Rightarrow n/m = k \Rightarrow m \text{ is a divisor of } n$$

$$\Rightarrow O(H) \text{ is divisor of } O(G)$$

$$\Rightarrow O(H) \text{ divides } O(G).$$

Hence proof.

Note

- The converse of the Lagrange theorem is not true. That is, if m is a divisor of n , then it is not necessary that G must have a subgroup of order m . For example, the alternating group A_4 of degree 4 is of order 12. But there is no subgroup of A_4 of order 6, though 6 is a divisor of 12
- It is evident from Lagrange's theorem that if G is a finite group of order n and if m is not a divisor of n , then can be no subgroup of G of order m .